



Interior Display System Proposal

Powered by CoachMate

Prepared for: **[Employer Name]**

Prepared by: **John**

Date: **02 August 2026**

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1. Executive Summary

This proposal, powered by CoachMate, introduces a modern interior passenger information display system designed specifically for older fleet vehicles such as the Volvo B7RLE. The system uses a controller-driven HDMI display, bypassing legacy DG3/IBIS hardware entirely. This delivers a TFT-grade passenger experience using readily available UK components, with minimal electrical integration and full control over the UI.

The solution is:

- **Modern** — full graphics, animations, branding
- **Compatible** — works with any older bus
- **Maintainable** — uses standard components
- **Cost-effective** — avoids EG4/Ethernet retrofits
- **Deterministic** — your controller drives all logic

2. System Architecture Overview

High-Level Architecture

24V Bus Power --> DC Regulators --> Bus Controller --> HDMI --> Interior Display

Application Flow

Operations PWA Creates routes, duties, schedules. Sends duty link to driver.

Driver PWA Receives duty link. If none: shows Route Selector. On Start: begins GPS polling and progress tracking. Sends structured progress events to Bus Controller.

Bus Controller Receives progress events. Renders final UI. Outputs HDMI to interior display.

Interior Display Shows modern passenger information UI.

3. Driver PWA (Dedicated Chapter)

The heart of the platform's real-time intelligence

The Driver PWA is the operational core of the ecosystem. It is responsible for:

- Duty retrieval
- Route selection
- GPS polling
- Stop detection
- Progress tracking
- Sending structured progress events to the Bus Controller

It runs on a dedicated mobile device mounted in the driver's cab.

3.1 Driver PWA Responsibilities

Duty Handling

- Receives duty link from Operations PWA
- Displays duty summary
- If no duty: presents Route Selector
- Ensures driver starts correct route

GPS Polling

- Polls GPS at fixed intervals
- Calculates distance to next stop
- Determines stop arrival/leave events
- Detects terminus arrival

Progress Event Generation

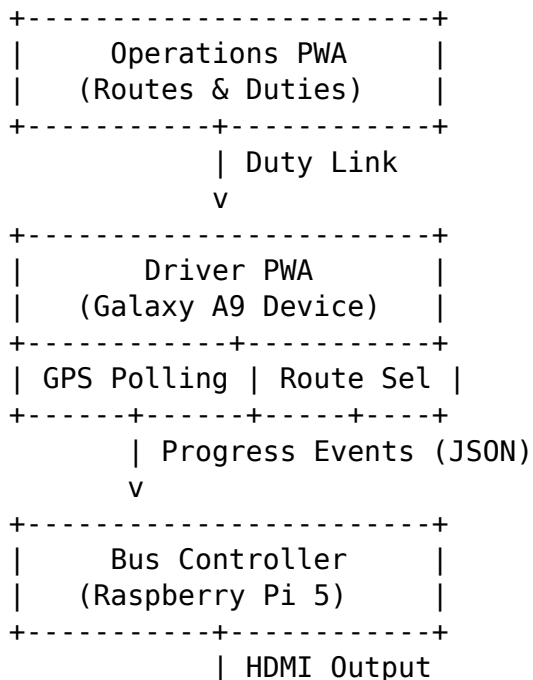
Driver PWA sends structured JSON events to the Bus Controller, including:

- next_stop
- following_stop
- distance_to_next
- eta_next_stop
- progress_percent
- stop_requested
- disruption_message
- terminus_flag

Connectivity

- WiFi hotspot
- Local network
- USB tether (optional)

3.2 Driver PWA Data Flow Diagram



V

```

+-----+
|   Interior Display   |
|   (Industrial LCD)   |
+-----+

```

4. Refined UI Mockups (Pixel-Level Specification)

4.1 Layout Zones

Zone A — Header Bar (Top 12%)

- Background: #0D0D0D
- Route number: Roboto Bold, 72px, #F9A825
- Destination: Roboto Medium, 48px, #FFFFFF
- Time: Roboto Regular, 36px, #CCCCCC
- Operator logo: white silhouette
- Company logo placed top-left in header

Zone B — Main Content (Middle 65%)

Next Stop (Primary)

- Font: Roboto Black, 96px, #FFFFFF
- Alignment: centred
- Animation: slide-in (300ms)

Following Stop (Secondary)

- Font: Roboto Regular, 48px, #AAAAAA
- Alignment: centred
- Animation: fade-in (200ms)

Icon Row

- Icons: wheelchair, bell, interchange
- Style: line icons, 2px stroke
- Colour: #F9A825
- Size: 48px

Zone C — Footer Bar (Bottom 23%)

Progress Bar

- Background: #1A1A1A
- Bar colour: #F9A825
- Height: 24px
- Animation: smooth linear progression

Disruption / Notices

- Font: Roboto Medium, 36px, #FF5252
- Behaviour: static or scrolling

4.2 Colour Palette

Purpose	Colour
Background	#0D0D0D
Primary text	#FFFFFF
Secondary text	#AAAAAA
Accent	#F9A825
Warning	#FF5252

4.3 Typography

- Roboto family (Bold, Medium, Regular, Black)
- High readability
- Clean, modern, widely supported
- Works well on LCD panels

5. Wiring Diagram (Text-Based)

Power Path

24V Bus Supply

```
| -- Fuse 10A --> 24V-to-12V Converter --> LCD Panel + Driver Board
`-- Fuse 5A  --> 24V-to-5V Converter  --> Raspberry Pi 5 (Bus Controller)
```

Signal Path

```
Driver PWA --> Progress Events --> Bus Controller
Bus Controller HDMI Out --> Shielded HDMI Cable --> HDMI Driver Board --
> LVDS --> LCD Panel
```

Grounding

```
LCD GND
Driver Board GND
Pi GND
DC Converter GND
--> Single Chassis Ground Point
```

6. Installation Plan (Volvo B7RLE)

6.1 Mechanical Installation

1. Identify ceiling cross-members above aisle.
2. Install mounting rails with M6/M8 bolts.
3. Add vibration dampers.
4. Mount enclosure flush to ceiling.
5. Install LCD panel and driver board inside enclosure.
6. Install Raspberry Pi in ventilated metal case.

6.2 Electrical Installation

1. Tap into auxiliary 24V circuit.
2. Install 10A and 5A fuses.
3. Mount DC-DC converters on metal plate.
4. Run 12V to LCD and driver board.
5. Run 5V to Raspberry Pi.
6. Install ferrite chokes on power lines.
7. Run HDMI cable from Pi to driver board.

6.3 Software Commissioning

1. Install Raspberry Pi OS Lite.
2. Configure auto-boot systemd service.
3. Set HDMI resolution to LCD native.
4. Test UI transitions and brightness.
5. Perform road test for readability and stability.

7. Procurement Tables (All Devices)

7.1 Driver PWA Device

Item	Model	Supplier	Notes
Driver PWA Device	Samsung Galaxy A9 (8GB)	Amazon UK / Samsung UK	Recommended
Rugged Case	SUPCASE UB Pro	Amazon UK	Shock-resistant
Vehicle Charger	24V to USB-C PD	RS / Amazon UK	Required for cab power

7.2 Bus Controller

Item	Model	Supplier	Notes
Bus Controller	Raspberry Pi 5 (8GB)	Amazon UK	HDMI output
Storage	NVMe or SD	Pi Hut	Fast boot
Case	Metal ventilated	Pi Hut	Thermal stability

7.3 Interior Display

Item	Model	Supplier	Notes
LCD Panel	Industrial 15-21"	DisplayTechnology	LVDS input
HDMI Driver Board	Generic HDMI-to-LVDS	Pi Hut	Required
Enclosure	Custom aluminium	Fractory	Powder-coated

7.4 Power System

Item	Model	Supplier	Notes
24-to-12V Converter	30A waterproof	Amazon UK	LCD power
24-to-5V Converter	DC-DC module	RS / Farnell	Pi power
Fuses	5A + 10A	Vehicle Wiring Products	Inline holders

7.5 Cabling & Mounting

Item	Model	Supplier	Notes
HDMI Cable	Shielded short	Pi Hut	Reduce interference
Automotive Wiring	24V rated	VWP	Power loom
Mounting Hardware	M6/M8 bolts	Screwfix	Ceiling rails

8. Project Timeline

Phase	Duration
Procurement	1-2 weeks
Enclosure fabrication	1 week
Installation (per bus)	3-5 hours
Software commissioning	1 hour
Road testing	1 day

9. Cost Summary

Category	Cost
Hardware	£250-£450
Enclosure	£60-£150
Installation	£150-£300
Total (per bus)	£460-£900

10. Benefits to the Operator

- Modern passenger experience
- Full control over UI and branding
- No dependency on DG3/IBIS
- No expensive EG4/Ethernet retrofit
- Uses standard UK-available components
- Easy maintenance and replacement
- Scalable across entire fleet

11. Conclusion

This system provides a modern, reliable, and cost-effective interior display for older buses. It leverages your existing software, standard UK components, and a clean HDMI architecture to deliver a premium passenger experience without the complexity of legacy bus systems.

12. PDF-Ready Structure

Use this structure when exporting:

1. Cover Page with company logo
2. Executive Summary
3. System Architecture
4. Driver PWA Chapter
5. UI Mockups
6. Data-Flow Diagrams
7. Wiring Diagram
8. Installation Plan
9. Procurement Tables
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